Pathophysiology of the digestive system

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GIT: Content

- Symptoms
  - Indigestion or dyspepsia or an upset stomach is a symptom (e.g. abdominal pain and a feeling of fullness soon after you start eating) of various digestive diseases, rather than a specific disease.
  - Dysphagia is the swallowing problems
  - Odynophagia is the painful swallowing
  - Pyrosis or heartburn is a burning sensation in the upper abdomen
  - Vomiting
  - Constipation
  - Diarrhea
  - Bleeding
  - Abdominal pain
- The diverticula: types
  - The pharynx
    - Zenker's diverticulum or systemic diseases: Sjögren syndrom
  - The esophagus
    - Achalasia and epiphrenic / supradiafragmal diverticula
    - Parabronchial diverticula
- The stomach
  - Gastroesophageal Reflux Disease
  - Hiatal hernia
  - Pyloric obstruction
  - Gastritis
  - Peptic Ulcer Disease: Duodenal ulcers and Gastric ulcers
  - Stress-Related Mucosal Disease and Dumping Syndrome
- The small Intestinal Obstruction and Ileus
- The Large intestine
  - Inflammatory Bowel Disease: Ulcerative Colitis (distal small intestine) and Crohn Disease
  - Diverticular Disease of the Colon
  - Irritable Bowel Syndrome
  - Megacolon



Signs and symptoms: content

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- Vomiting
- Constipation
- Diarrhea
- Bleeding
- Abdominal pain



Vomiting

- the duodenum and antrum of the stomach produce retrograde peristalsis while the body of the stomach and the esophagus relax
- deep inspiration and the glottis closes
- the diaphragm is forced high into the thoracic cavity by strong contractions of the abdominal muscles
- the higher intrathoracic pressure forces the upper esophageal sphincter to open, and chyme is expelled from the mouth
- the stomach relaxes and the upper part of the esophagus contracts, forcing the remaining chyme back into the stomach

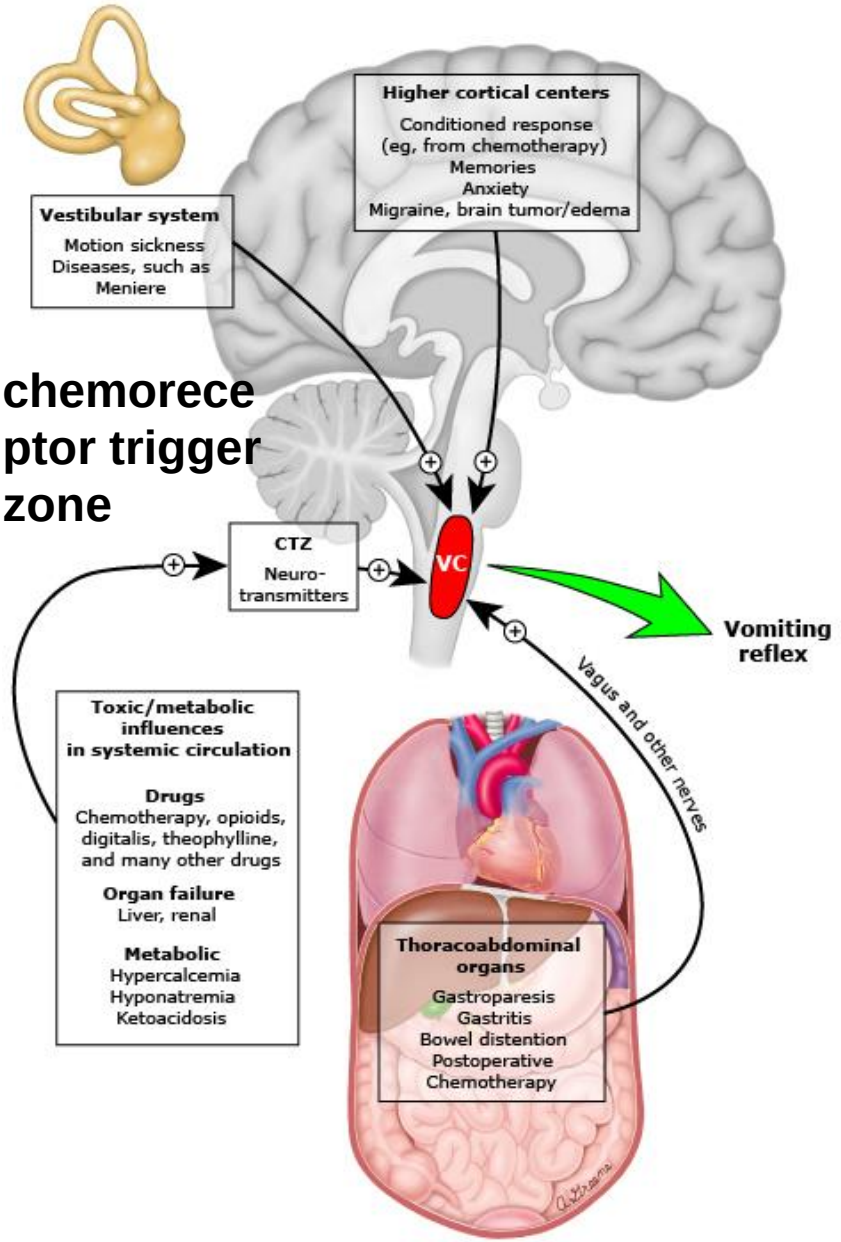
A sympathetic discharge causes the tachycardia, tachypnea, and sweating.
The parasympathetic system mediates copious salivation, increased gastric motility, and relaxation of the upper and lower esophageal sphincters.

Vomiting with/without nausea

- Nausea and retching usually precede vomiting
  - **Nausea** is a subjective experience
  - **Retching** is the muscular event of vomiting but the contents of the esophagus drop back into the stomach as the upper esophageal sphincter is closed
- **Projectile vomiting** is spontaneous vomiting that is not preceded by nausea or retching. Projectile vomiting is caused by direct stimulation of the vomiting center by neurologic lesions

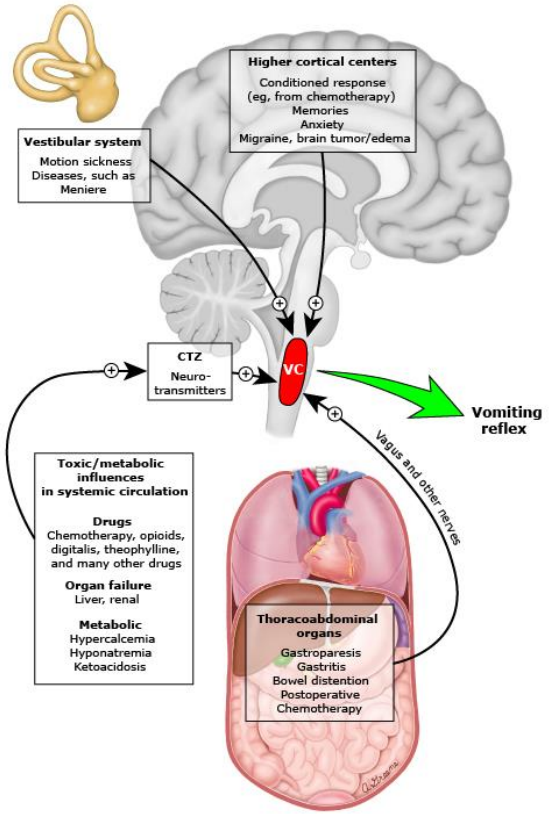
The vomiting center

- The vomiting center lies in the medulla oblongata and includes the reticular formation. The stimuli arising from:
  - the nucleus tractus solitarius
  - the cortex and thalamus (anxiety and pain)
- The nucleus tractus solitarius integrates inputs noxious stimuli arising from
    - the vestibular system through the vestibulocochlear nerve (VIII, motion sickness)
    - the chemoreceptor trigger zone



Vomiting: chemoreceptor trigger zone

- The chemoreceptor trigger zone (the area postrema = a sensory circumventricular organ without a blood–brain barrier) has the receptors for dopamine, serotonin, histamine, substance P, opioid and acetylcholine receptors.
  - These receptors monitor blood content and signal to the vomiting center.
  - They monitor intestinal, vagal, sympathetic input, including: pharyngeal stimulation, myocardial ischemia, gastric irritation or distension.
  - Serotonin is produced by the intestinal enterochromaffin cells which can be stimulated by luminal distension or changed osmotic concentrations in intestinal contents.
    - Serotonin antagonists are effective antiemetics (drugs preventing or relieving nausea and vomiting) and have been used to treat nausea and vomiting associated with postoperative vomiting and cancer chemotherapy



Constipation

difficult or infrequent defecation

- Primary constipation
  - *Normal transit (functional) constipation:* a normal stool passage but difficulty with stool evacuation. Causes: low-residue, low fluid intake, low physical activity
  - *Slow-transit constipation:* impaired colonic motor activity with infrequent bowel movements, straining to defecate, mild abdominal distention, and palpable stool in the sigmoid colon.
  - *Pelvic floor dysfunction:* difficulty expelling stool because of failure of the pelvic floor muscles or anal sphincter to relax with defecation.
- Secondary constipation

Neurogenic disorders: stroke, Parkinson disease, spinal cord injury, multiple sclerosis

Endocrine or metabolic disorders: hypothyroidism, diabetes mellitus, hypokalemia, and hypercalcemia.

Pelvic hiatal hernia (herniation of the bowel through the floor of the pelvis), diverticuli, irritable bowel syndrome– constipation predominant,

Drugs: codeine, antacids containing calcium carbonate

Pregnancy

Aging due to degeneration of neurons in the myenteric plexus

Diarrhea:

- **Large-volume diarrhea:**
  - **Osmotic diarrhea:** a nonabsorbable substance in the intestine draws water into the lumen by osmosis. Causes: oral doses of magnesium, sulfate, and phosphate, synthetic sugars: sorbitol) or lactase deficiency, pancreatic enzyme or bile salt deficiency, small intestine bacterial overgrowth, and celiac disease.
  - **Secretory diarrhea** caused by excessive mucosal secretion of chloride- or bicarbonate-rich fluid or inhibition of net sodium absorption. Infectious causes include viruses (rotavirus), bacterial enterotoxins (*Escherichia coli* and *Vibrio cholerae*)
- **Small-volume diarrhea** usually is caused by an inflammatory disorder of the intestine, such as ulcerative colitis, Crohn disease, or microscopic colitis
- **Motility diarrhea:** excessive motility caused by resection of the small intestine (short bowel syndrome), surgical bypass of an area of the intestine, fistula formation between loops of intestine, irritable bowel syndrome–diarrhea predominant, diabetic neuropathy, hyperthyroidism, and laxative abuse.

Abdominal pain

- **Parietal (somatic) pain** arises from the parietal peritoneum.
    - More intense
    - More localized : Nerve fibers from the parietal peritoneum travel with peripheral nerves to the spinal cord, and the sensation of pain corresponds to skin dermatomes
- **Visceral pain** arises from a stimulus (distention, inflammation, ischemia) acting on abdominal organ
    - An dull (vague) rather than sharp, and is difficult to describe
    - Poorly localized (diffuse):Pain is usually felt near the midline in the epigastrium, midabdomen, or lower abdomen.
    - **Referred pain** is visceral pain felt at some distance from a diseased or an affected organ. It is felt in the skin dermatomes or deeper tissues that share a central afferent pathway with the affected organ. Gallbladder pain is, for example, referred to the right shoulder or scapulae.

Gastrointestinal (GI) bleeding

- **Upper GI bleeding** is bleeding in the esophagus, stomach, or duodenum and is characterized by
  - **Hematemesis:** is the vomiting of blood, which may be obviously red or have an appearance similar to coffee grounds.
  - **Melena**: dark, grainy digested blood (“coffeegrounds”) in stool.
- **Lower GI bleeding:** bleeding from the small intestine (jejunum or ileum), colon, or rectum (polyps, inflammatory bowel disease, diverticulosis, cancer, or hemorrhoids)
  - **Haematochezia** (enterorrhagia)**:** fresh blood in stool (ulcerative colitis, in the ileum and transverse colon or hemorrhoids)
    - Haematochezia from an upper gastrointestinal source is an ominous sign, as it suggests a very significant bleed which is more likely to be life-threatening
  - **Rectorrhagia**: rectal bleeding without defecation; expulsion of fresh bright red blood
- **Occult bleeding** is usually caused by slow, chronic blood loss that is not obvious and results in iron deficiency anemia as iron stores in the bone marrow are slowly depleted.
- **Acute, severe GI bleeding** is life threatening depending on the volume and rate of blood loss, associated disease, age, and effectiveness of treatment.

Dysphagia: oropharyngeal

- It is difficulty swallowing. Disorders that affect the striated muscles of the upper esophagus interfere with the oropharyngeal (voluntary) phase of swallowing.
  - Mechanical obstructions:
    - Intrinsic: originate in the wall of the esophageal lumen(tumors, strictures, and diverticular herniation, see diverticula)
    - Extrinsic: originate outside the esophageal lumen and narrow the esophagus by pressing inward on the esophageal wall.(tumor)
  - Functional dysphagia:
    - Systemic disorders:
      - Sjogren syndrome
      - Dermatomyositis (an idiopathic inflammatory symmetric, proximal skeletal muscle disease)
      - Neurologic impairments: caused by
        - Stroke
        - Multiple sclerosis
        - Parkinson disease
        - Amyotrophic lateral sclerosis
        - Myasthenia gravis
        - Poliomyelitis (poliovirus)

Dysphagia: Sjögren syndrome or Sjögren's syndrome

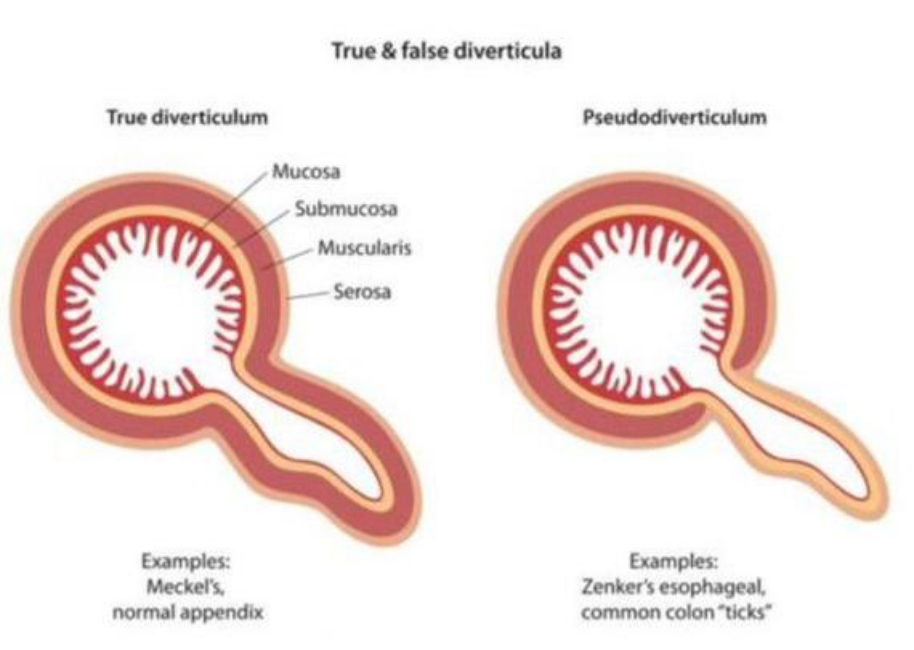
- A long-term autoimmune disease leading to loss of exocrine glands: salivary glands and often seriously affects other organ systems, such as the lungs, kidneys, and nervous system
- The hallmark symptom of Sjögren syndrome is dry mouth (xerostomia) and xerophthalmia (dry eyes) with keratoconjunctivitis. Sicca syndrome refers to dry eyes and mouths.
- It is the second-most common rheumatic autoimmune disorder, behind rheumatoid arthritis and systemic lupus erythematosus

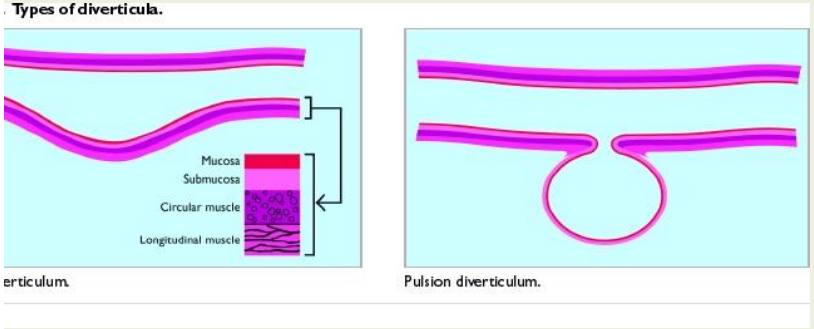
The esophagus: content

- Achalasia and diverticula (types and examples): affected motility
- Gastroesophageal reflux: lesion
- Hiatal hernia: types
- Mallory–Weiss syndrome
- Systemic diseases: e.g. sclerodermia

Types of diverticula

- Traction diverticula: resulting from a pulling force on the outside of the alimentary tract, are more commonly seen in the mid- to distal esophagus and often result from inflammatory or neoplastic processes adjacent to the muscularis.
- Pulsion diverticula: result from an internal pushing force (intraluminal pressure) causing the mucosa and submucosa to herniate through the muscular wall. These are pseudodiverticula: mucosa and submucosa through the muscularis propria.





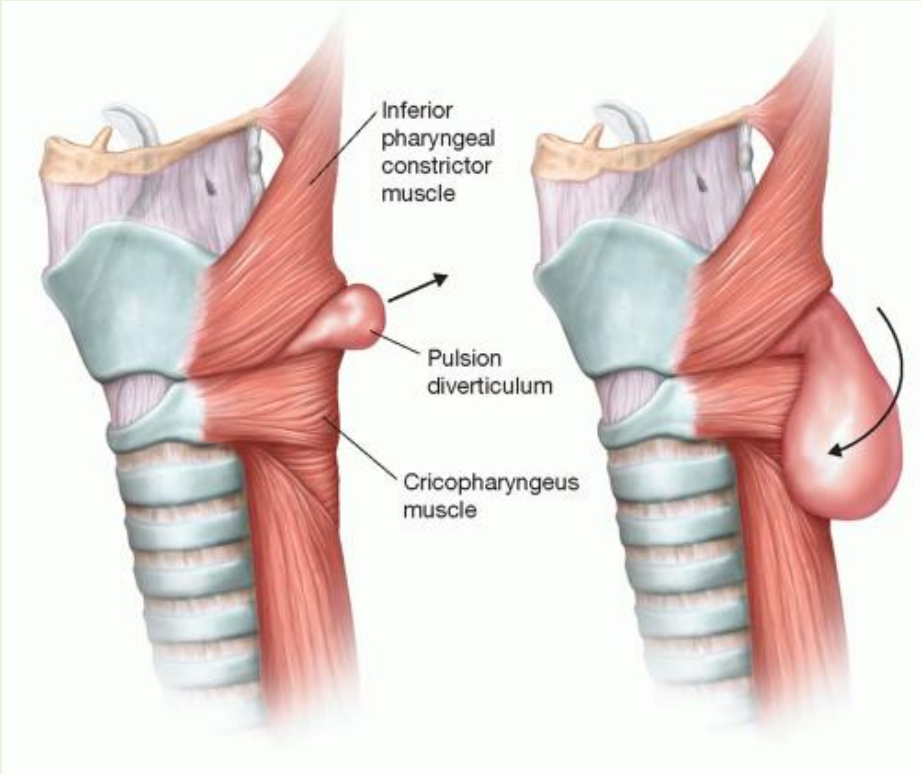
Contain only mucosa and submucosa

Pharynx: Zenker's diverticulum or pharyngeal pouch

- Diverticulum of the mucosa of the human pharynx, just above the cricopharyngeal muscle (above the upper sphincter of the esophagus).
- Pseudo diverticulum: not involving all layers of the esophageal wall
- Pulsion diverticula: result from an internal pushing

Clinical symptoms:

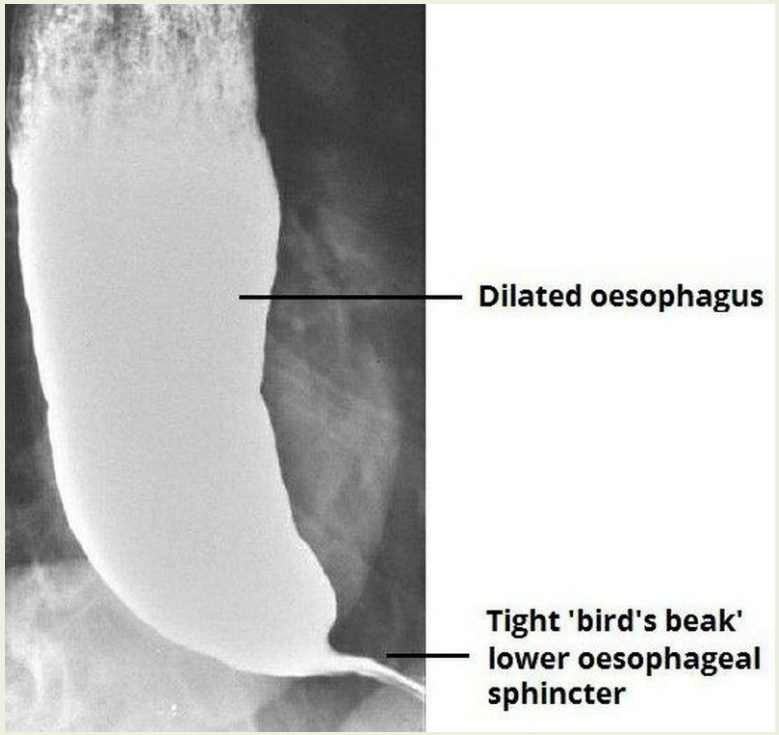
- difficulty swallowing (dysphagia)
- spontaneous return of undigested food (regurgitation of undigested food)
- chronic aspiration and frequent respiratory infections (aspiration pneumonia)
- bad breath (fetor ex-ore) due to the existence of decomposed food ingredients in a diverticulum.



The esophagus: Achalasia

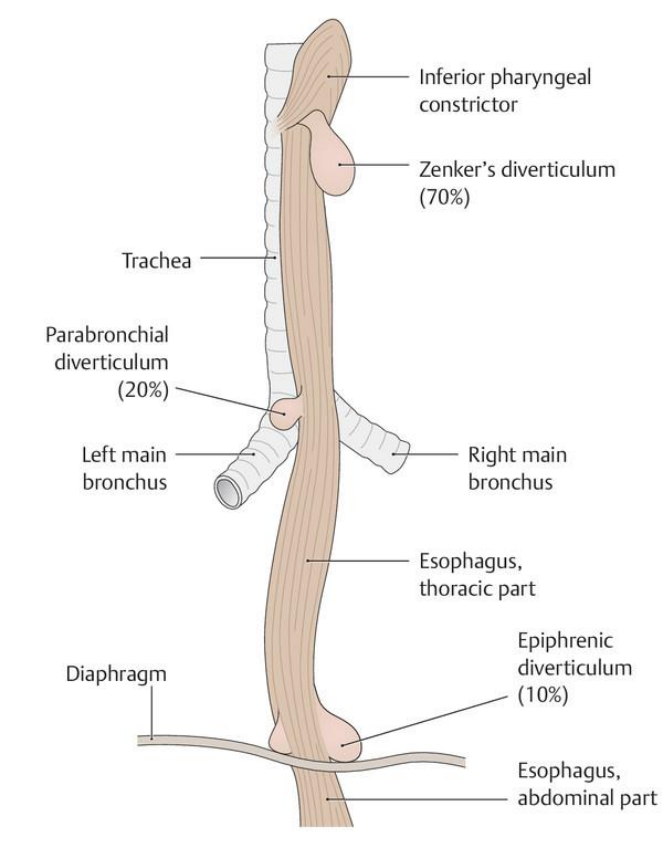
- a rare disorder related to loss of inhibitory neurons in the myenteric plexus with smooth muscle atrophy in the middle and lower portions of the esophagus. This leads to altered esophageal peristalsis and failure of the lower esophageal sphincter to relax.

Unknown etiology: viral or autoimmune mechanisms. This leads to altered esophageal peristalsis and failure of the lower esophageal sphincter to relax



Epiphrenic or supradiafragmal diverticula

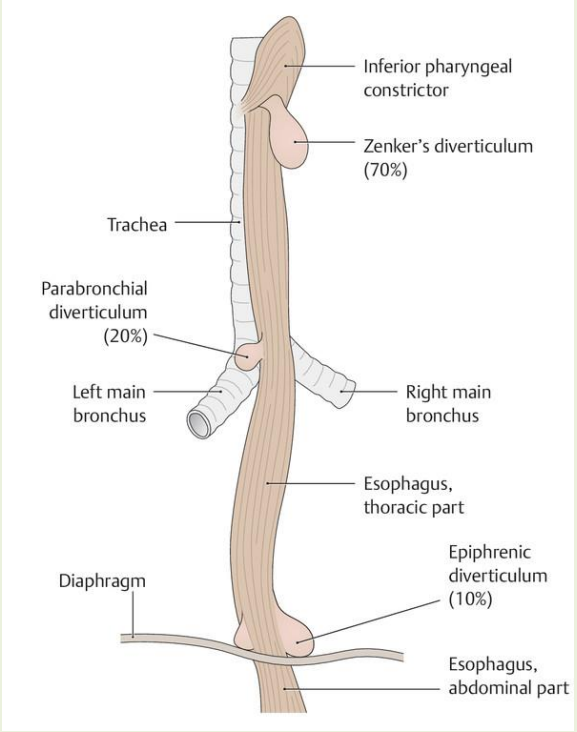
- on the body of the esophagus and above the diaphragm
- The predominant motor disorder is diffuse spasm or functional obstruction usually done on patients primarily suffering from achalasia.
- Pseudo diverticulum: not involving all layers of the esophageal wall
- Pulsion diverticula



Clinical manifestation:
Dysphagia
Regurgitation
Retrosternal pain mostly
due to the joint nutcracker esophagus

Parabronchial diverticulum: in the middle of the chest

- Traction diverticula are usually localized in the thoracic esophagus (tracheal bifurcation).
- Results from inflammation of lymph nodes with subsequent scarring and adhesion of matted nodes to the esophagus. (tuberculosis or histoplasmosis).
- Traction diverticula
- True diverticula



Gastroesophageal reflux leading to Gastroesophageal Reflux Disease

Gastroesophageal Reflux Disease: the regurgitation of acid and pepsin from the stomach to the esophagus that causes **esophagitis.**

- *Physiologic reflux:* Gastroesophageal reflux that does not cause symptoms
- *Nonerosive reflux disease:* symptoms of reflux disease but no visible esophageal mucosal injury

Causes and risk factors: weaken low esophageal sphincter (LES) mediated by

- Hiatal hernia
- Drugs or chemicals that relax the LES: anticholinergics, nitrates, calcium channel blockers, nicotine
- Increases abdominal pressure: vomiting, coughing, lifting, bending, or obesity or asthma or chronic cough

Later the inflammation can further damage vagus innervation that lead to delay gastric emptying worsening motility: lengthened period during which reflux is possible or increasing the acid content of chyme, e.g. gastric or duodenal ulcers, which can cause pyloric edema

Gastroesophageal reflux leading to Gastroesophageal Reflux Disease

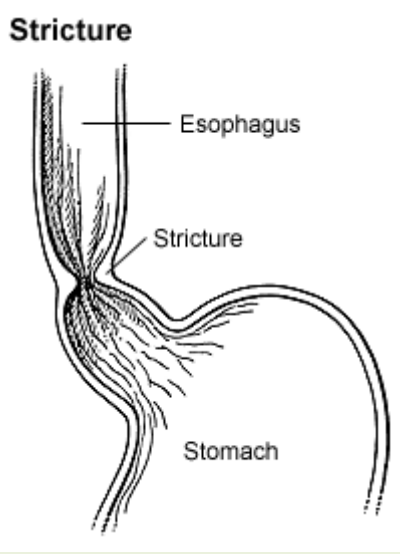
Pathophysiology:

transient relaxation or weakness or low resting tone of the low esophageal sphincter
Inflammatory responses in the esophageal wall resulting in: hyperemia, edema, tissue fragility, erosion, and ulcerations, fibrosis (strictures or abnormal narrowing), basal cell hyperplasia, and elongation of papillae.
Barrett esophagus: precancerous lesions can progress to adenocarcinoma

The clinical manifestations:

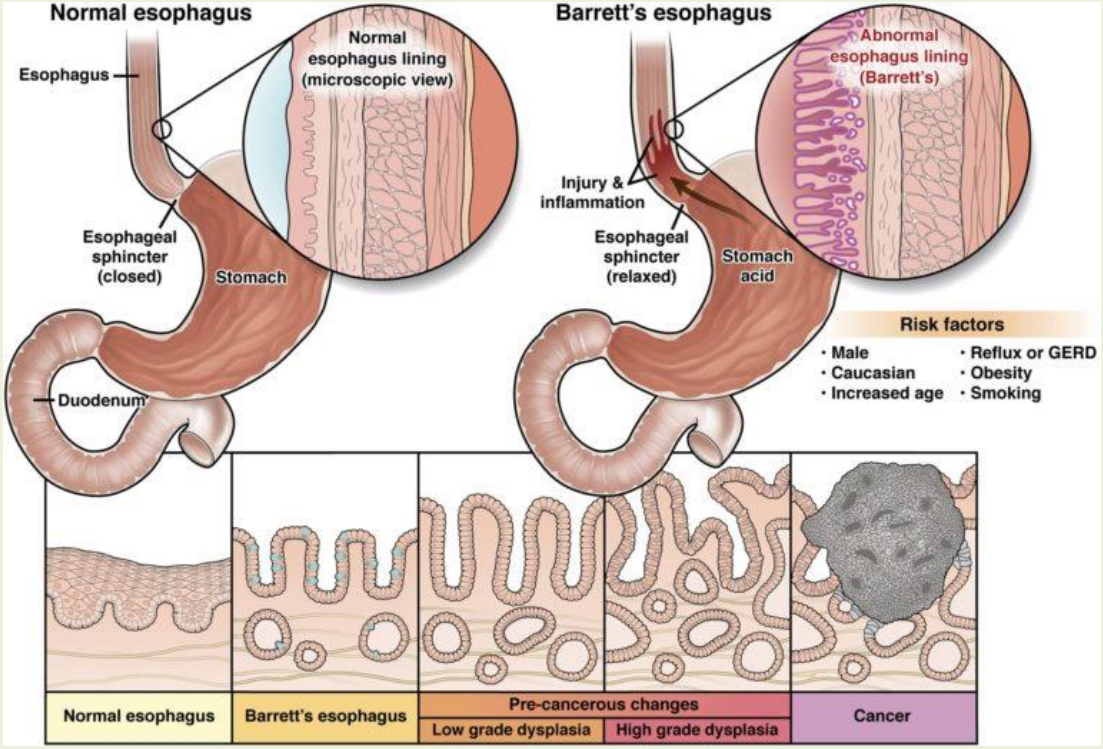
- Heartburn from acid regurgitation: Heartburn also may be experienced as chest pain, which requires ruling out cardiac ischemia.
- Chronic cough, asthma attacks
- Laryngitis
- Dysphagia.

Upper abdominal pain usually occurs within 1 hour of eating and can be relapsing and remitting. Symptoms can worsen if the individual lies down.

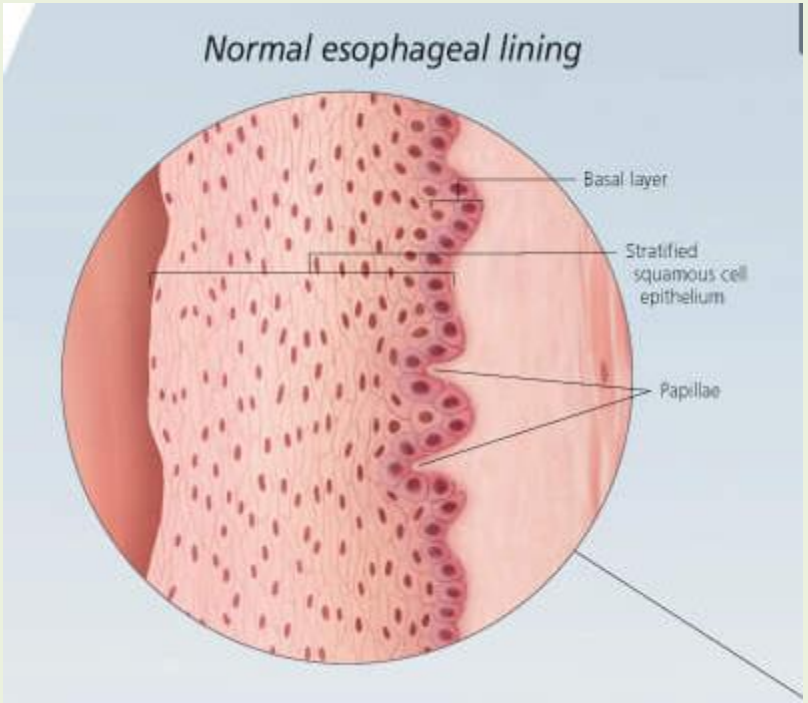


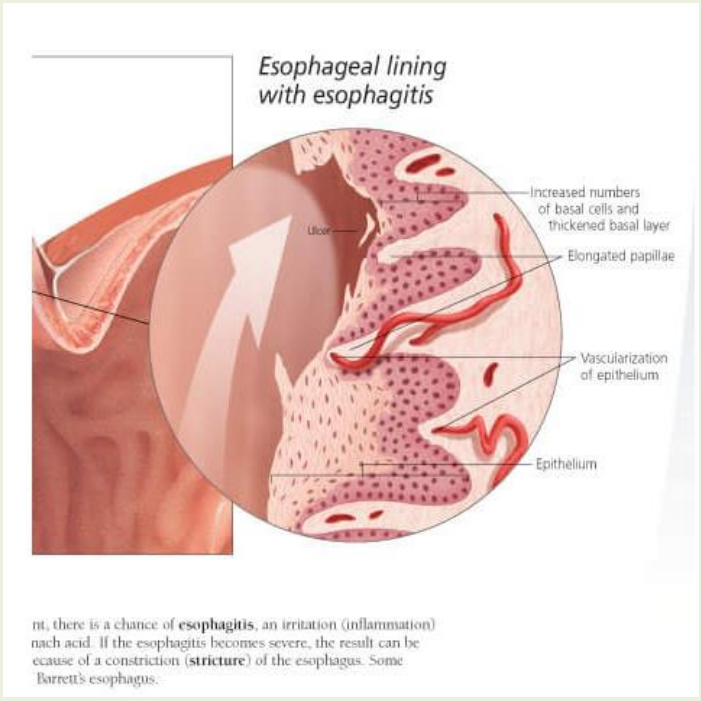
Barrett esophagus

A premalignant condition because it can lead to transition to esophageal adenocarcinoma.

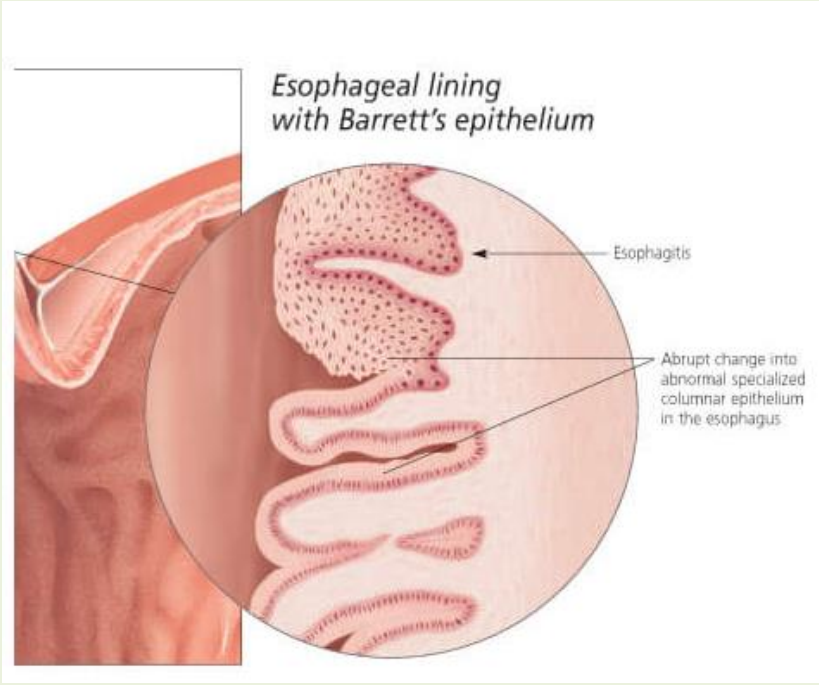


change in the mucosal cells lining the lower portion of the esophagus, from stratified squamous epithelium to simple columnar epithelium with interspersed goblet cells that normally secrete mucins only in the small intestine and large intestine.





nt, there is a chance of **esophagitis**, an irritation (inflammation) nach acid. If the esophagitis becomes severe, the result can be ecause of a constriction (**stricture**) of the esophagus. Some Barrett's esophagus.



Hiatal hernia

- The protrusion (herniation) of the upper part of the stomach through the diaphragm into the thorax.
- Four types:
  - Sliding (type I):
  - Paraesophageal (type II):
  - Mixed (type III): which include elements of types I and II
  - Type IV: the entire stomach and other abdominal organs slide into the thorax.

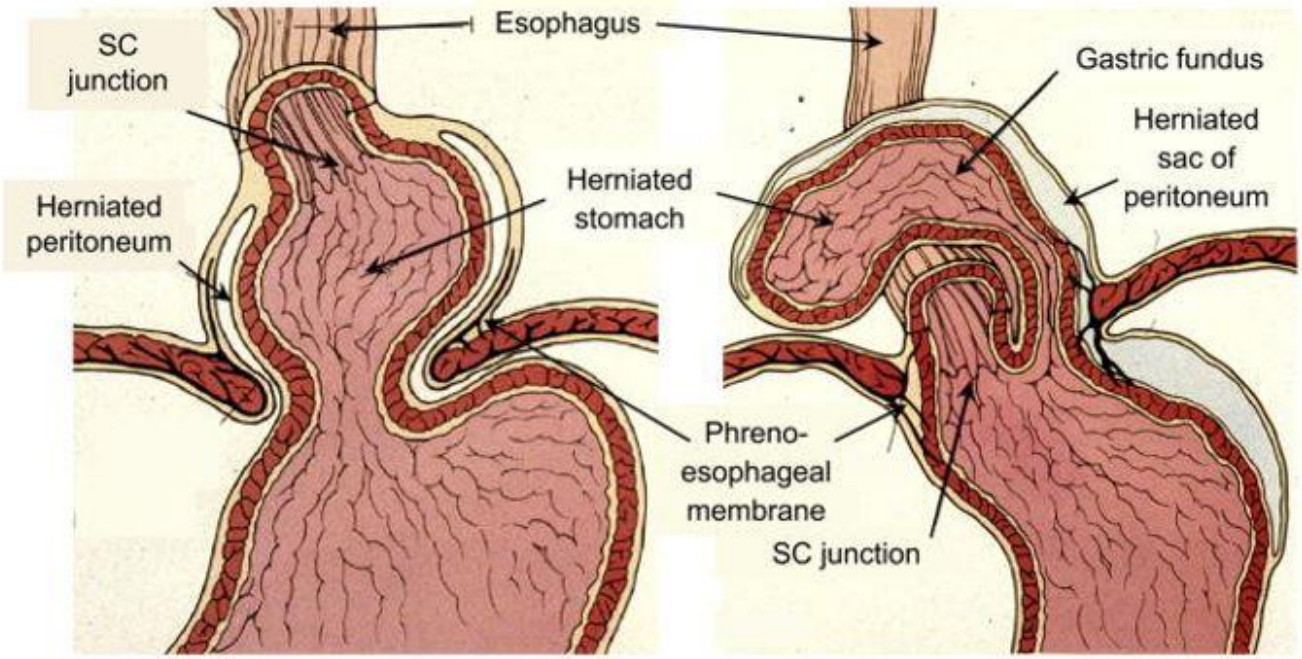
Hiatal hernia:
Type I or sliding hiatal hernia

- the most common type
- the proximal portion of the stomach moves into the thoracic cavity through the esophageal hiatus, an opening in the diaphragm for the esophagus and vagus nerves.
  - Supine position:, the lower esophagus and stomach are pulled into the thorax.
  - Standing causes the organs to “slide” back into the abdomen. Sliding hiatal hernia is by factors that
  - Symptoms are exacerbated by intra-abdominal pressure: coughing, bending, wearing tight clothing, ascites, obesity, or pregnancy (progesterone and estrogen may lower the resting pressure of the LES further).
- This type of hernia is associated with gastroesophageal reflux because the hernia diminishes the resting pressure of the LES.

Hiatal hernia:
Type II, paraesophageal hiatal hernia (rolling hiatal hernia)

- herniation of the greater curvature of the stomach is through a secondary opening in the diaphragm. As the stomach protrudes through the opening into the thorax, it lies alongside the esophagus.
- The gastroesophageal junction remains below the diaphragm.
- The position of a portion of the stomach above the diaphragm, however, causes congestion of mucosal blood flow and can lead to gastritis and ulcer formation. A mechanical strangulation of the hernia is a major complication, and surgical correction is required. Strangulation occludes blood vessels and causes vascular engorgement,edema, ischemia, and hemorrhage.

Type I, sliding hiatal hernia
vs
Type II, paraesophageal hiatal hernia (rolling hiatal hernia)

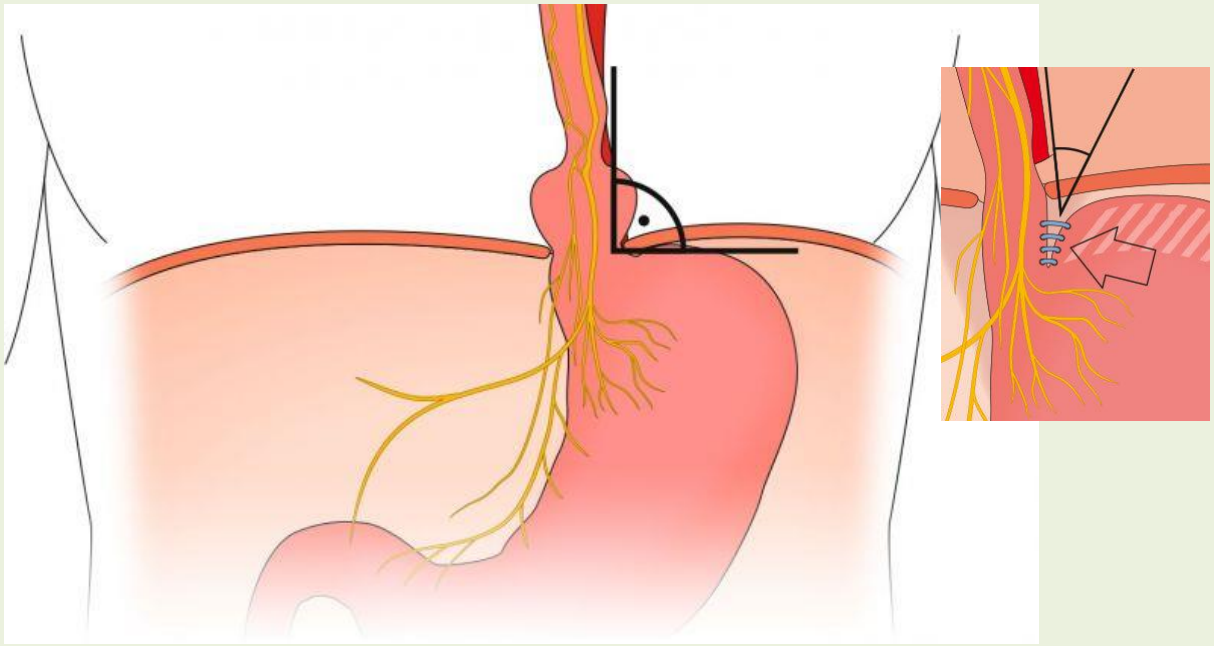


Distinction between a sliding hiatal hernia (type I) and paraesophageal hernia (type II). With type I hernia the leading edge is the gastric cardia while with type two it is the gastric fundus. The squamocolumnar junction (SCJ) maintains its native position in the paraesophageal hernia while it is displaced upward with the sliding hernia.

Best Pract Res Clin Gastroenterol. 2008; 22(4): 601–616.

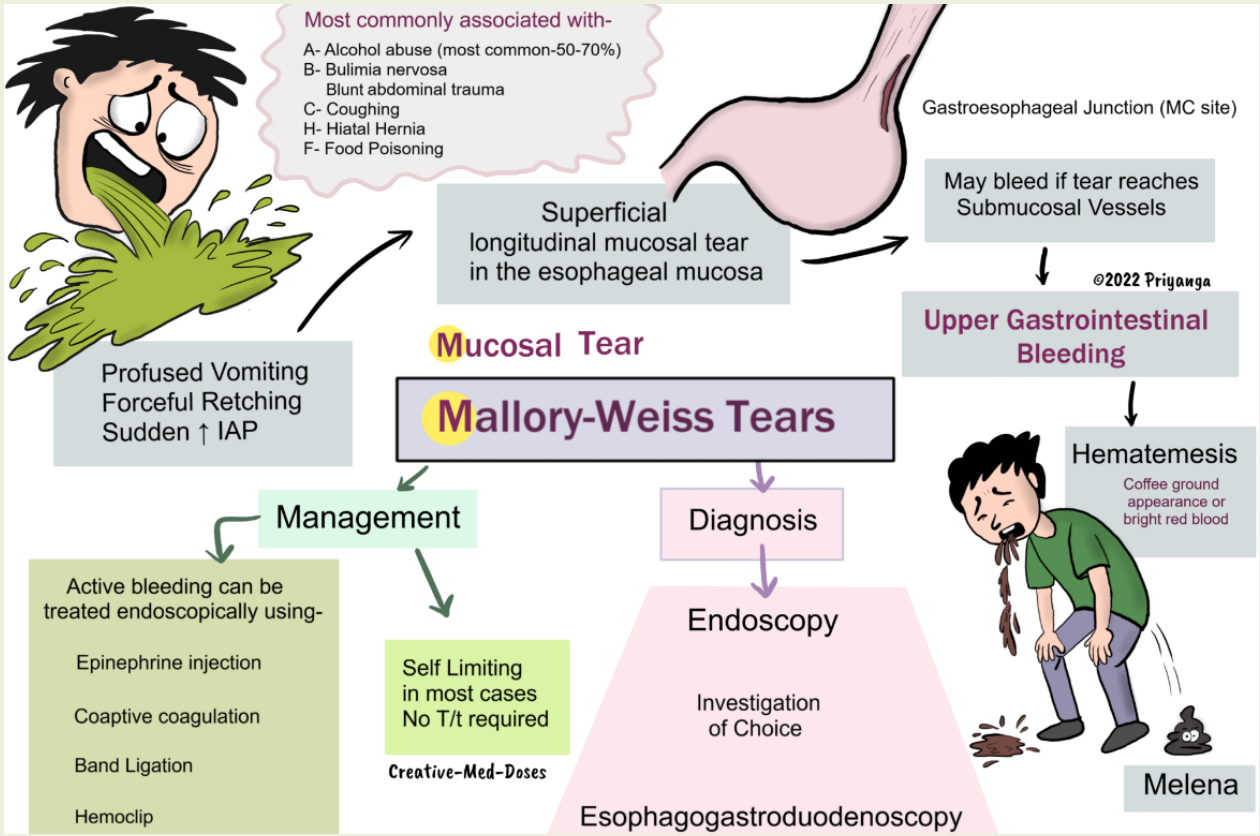
Type I, sliding hiatal hernia
vs
Type II, paraesophageal hiatal hernia (rolling hiatal hernia)

Type I: Absent angle of His (esophagogastric angle)
Type II: Present angle of His (esophagogastric angle)



Mallory–Weiss syndrome or gastro-esophageal laceration syndrome

- It refers to bleeding from a laceration in the mucosa at the junction of the stomach and esophagus. This is usually caused by severe vomiting because of alcoholism or bulimia, but can be caused by any condition which causes violent vomiting and retching such as food poisoning. The syndrome presents with hematemesis. The laceration is sometimes referred to as a Mallory–Weiss tear.



Pyloric obstruction (gastric outlet obstruction)

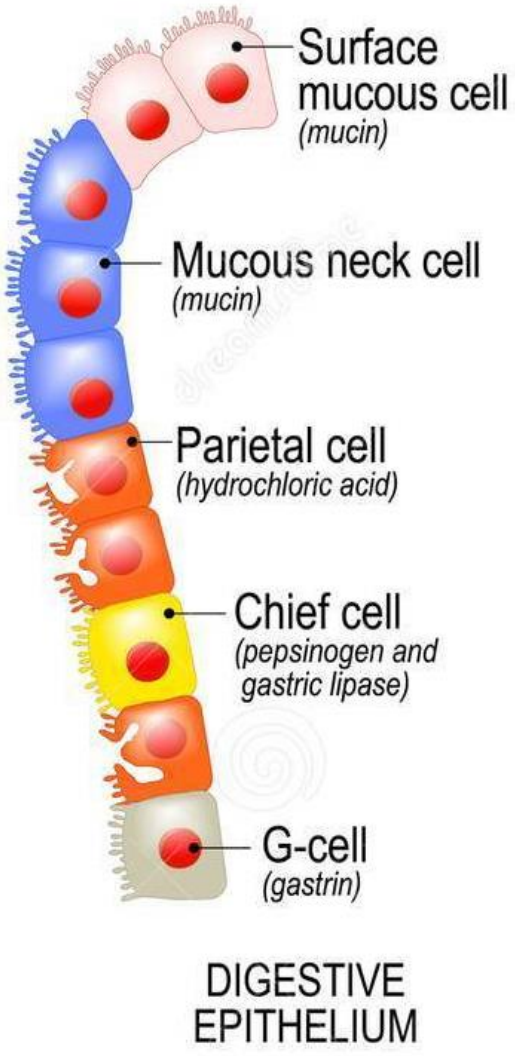
- The narrowing or blocking of the opening between the stomach and the duodenum.
  - Congenital: infantile hypertrophic pyloric stenosis
  - Acquired: peptic ulcer disease or carcinoma near the pylorus. Duodenal ulcers are more likely than gastric ulcers to obstruct the pylorus. Ulceration causes obstruction resulting from inflammation, edema, spasm, fibrosis, or scarring. Tumors cause obstruction by growing into the pylorus

- Symptoms:
  - Less severe obstruction causes vague epigastric fullness
  - Severe obstruction causes gastric distention and atony (lack of muscle tone and gastric motility). Cardinal signs: vomiting and rolling or jarring of the abdomen produces a sloshing sound called the succussion or gastric splash.

The Stomach

- Introduction: mucosa of stomach
- Zollinger-Ellison syndrome
- Gastritis
  - Acute and chronic (type A and B)
- Peptic Ulcer Disease
  - Gastric and duodenal ulcers
  - Defensive factors and aggressive factors
    - Helicobacter Pylori
  - Ischemic and curling ulcers
- Postprandial Dumping syndrome

Mucosa of stomach



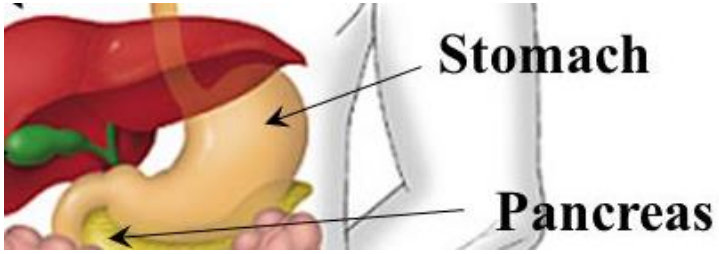
- **Parietal cells** secrete
  - hydrochloric acid
  - intrinsic factor (pernicious anemia)
  - gastroferrin (It facilitates small intestinal absorption of iron)

- **Chief cells** secrete:
  - pepsinogen which is readily converted to pepsin (proteolytic enzyme) at a pH of 2.
  - Lipase
  - HCO_3^-

- **G cells** secrete: gastrin which stimulates HCl and gastric motility (mainly pyloric gland mucosa in the antrum) and pancreatic juice**.**

- **Enterochromaffin-like cells** secrete histamine**.** Histamine receptors in the gastric mucosa are H2 receptors (unlike those in the bronchial mucosa, which are H1 receptors). They stimulates gastric acid secretion.
- **D cells** secrete somatostatin which inhibits acid secretion.

Zollinger-Ellison syndrome



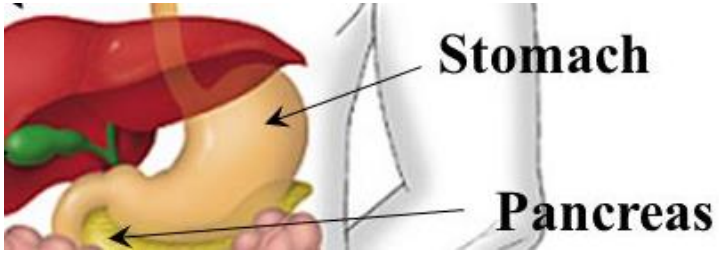
- A gastrinoma (benign or malignant tumors) of the pancreas or duodenum (a gastrin-secreting

neuroendocrine tumor or multiple tumors) stimulates a proliferation of gastric parietal cells and chronic secretion of gastric acid.

  - ↑gastrin, which causes ↑ gastric acid.

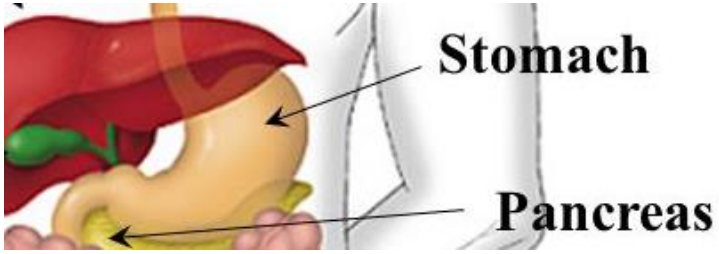
- It causes gastric and duodenal ulcers, gastroesophageal reflux

Gastritis



an inflammatory disorder of the gastric mucosa. The erosions are usually superficial.

Acute gastritis



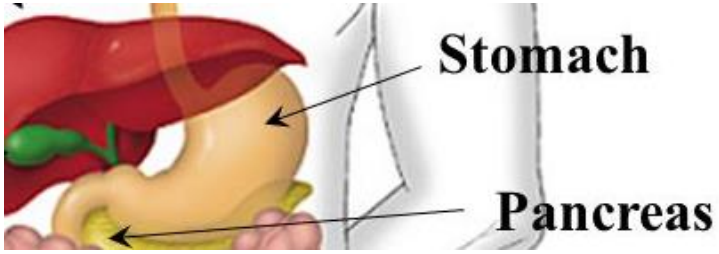
caused by injury of the protective mucosal barrier

- Drugs: Nonsteroidal anti-inflammatory drugs NSAIDs (ibuprofen or aspirin)

inhibit cyclooxygenase-1, and thus inhibit prostaglandins, which normally stimulate the secretion of mucus and suppress inflammation.

- *Helicobacter pylori:* causes
  - Inflammation
  - Down-regulation of antral somatostatin leading to increased gastrin, increased acid, impaired mucosal bicarbonate production, and increased mucosal exposure to acid and pepsin.
  - Secretion of urease that produces ammonia and carbon dioxide, resulting in a more alkaline environment
- Others: alcohol, histamine and metabolic disorders such as uremia are contributing factors.

Chronic gastritis

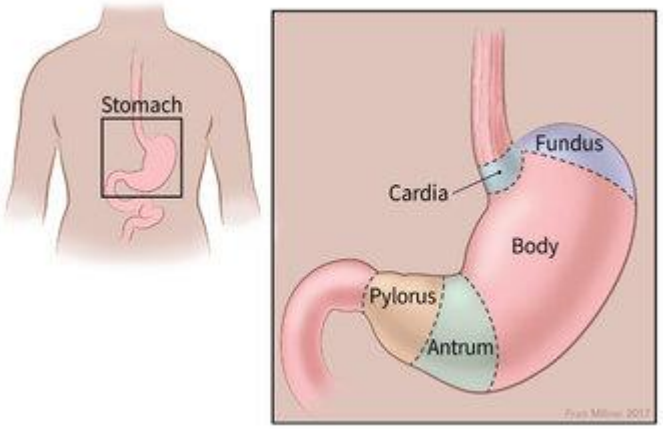


occur in older adults and causes chronic inflammation, mucosal atrophy, and epithelial metaplasia.

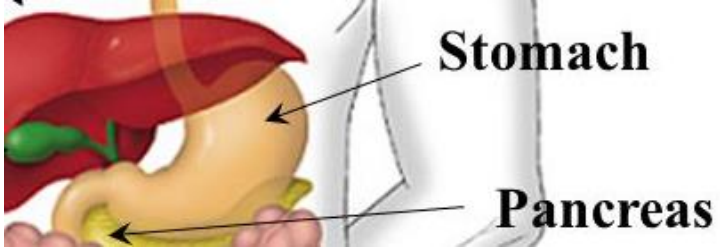
- type A, or immune (fundal): rare and severe type
- type B, nonimmune (antral): common

atrophic gastritis: antiparietal cell antibody and elevated plasma ghrelin level

H. pylori gastritis : *H. pylori* infection



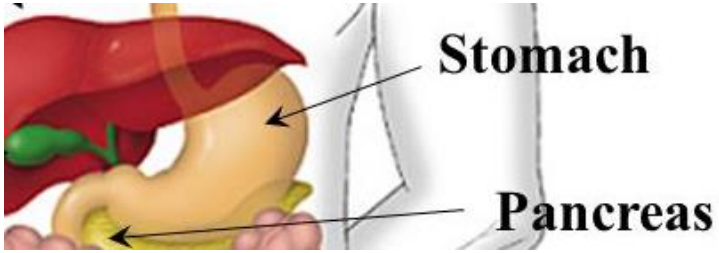
Chronic fundal gastritis: type A



loss of T-cell tolerance and development of autoantibodies to gastric H^+-K^+ ATPase.
H. pylori can trigger the immune response through molecular Mimicry

- Loss of parietal cells
  - diminishes secretion of hydrochloric acid (achlorhydria, diminished secretion of hydrochloric acid) causing elevated plasma levels of gastrin
  - decreased intrinsic factor causing pernicious anemia

Chronic antral gastritis: type B

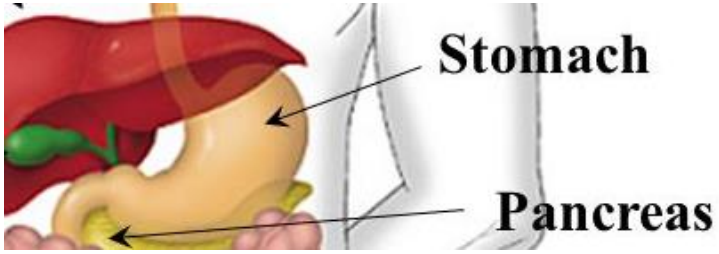


It is caused by

- *H. pylori* bacteria
- chronic use of alcohol, tobacco, and nonsteroidal antiinflammatory drugs.

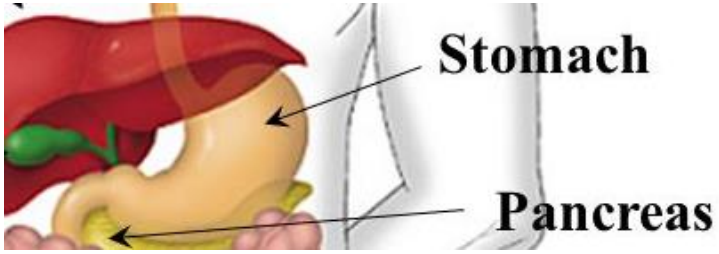
There are high levels of hydrochloric acid secretion with an increased risk of duodenal ulcers. *H. pylori* can also progress to autoimmune atrophic gastritis and involve the fundus, thus becoming pangastritis. In these cases there is greater risk for the development of gastric cancer

Alkaline reflux gastritis



- a stomach inflammation caused by reflux of bile and alkaline pancreatic secretions that contain proteolytic enzymes and disrupt the mucosal barrier in the remnant stomach.
- Clinical manifestations include nausea, bilious vomiting (vomiting in which the vomitus contains bile), and sustained epigastric pain that worsens after eating and is not relieved by antacids.

Peptic Ulcer Disease

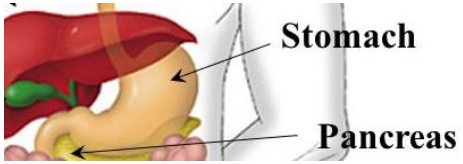


- *Erosions:* superficial ulcerations
- True ulcers: extend through the muscularis mucosae and damage blood vessels, causing hemorrhage or perforating the gastrointestinal wall.

- Gastric ulcers:
  - absent hyperacidity,
  - loss of weight,
  - 1:1 man : women and older age

- Duodenal ulcers:
  - greater frequency than other types of peptic ulcers,
  - hyperacidity
  - possible overweight

Gastric ulcers: pathophysiology



- An abnormality in balance between protective and aggressive factors (e.g. mucus, somatostatin from gastric D cells and others versus histamine, pepsin, NSAIDs, nicotine, alcohol, stress and others) that increases the mucosal barrier's permeability to hydrogen ions or in other words, decreased ability to withstand normal gastric acidity. Gastric hypoacidity is not typical feature.

  - frequently caused by *H. pylori* (an aggressive factor)
  - frequently due to duodenal reflux after cholecystectomy:
    - Alkaline reflux gastritis: Reflux of bile and pancreatic enzymes damage the gastric mucosa. The damaged mucosal barrier permits hydrogen ions to diffuse into the mucosa, where they disrupt permeability and cellular structure. A vicious cycle can be established as the damaged mucosa liberates histamine, which stimulates the increase of acid and pepsinogen production, blood flow, and capillary permeability. The disrupted mucosa becomes edematous and loses plasma proteins. Destruction of small vessels causes bleeding.

- Usually antral region, adjacent to the acid-secreting mucosa of the body

Duodenal ulcers: pathophysiology

- Pathophysiology: Hyperacidity: Gastric acid and pepsin penetrate duodenal mucosal barrier leading to ulceration

- Genetic factors, often between 20-40 years, increasing prevalence among women, 90-95% H pylori

- Manifestation:
  - chronic intermittent pain in the epigastric area when the stomach is empty. (The pain begins 30 minutes to 2 hours after eating, when the stomach is empty). It is not unusual for pain to occur in the middle of the night and disappear by morning (It can lead to overweight). The pain results from sensorineural stimulation by acid or muscle spasm, or both. Pain is relieved rapidly by food or antacids.
  - Bleeding from duodenal ulcers causes hematemesis or melena.

Peptic ulcer:
defensive factors and aggressive factors

Defensive factors:

- Compact epithelial lining: tight junctions
- Bicarbonate ions
- Mucus covering
- Prostaglandins and nitric oxide protect the mucosal barrier by stimulating the secretion of mucus and bicarbonate and by inhibiting the secretion of acid.
- Somatostatin, and secretin inhibit acid secretion.
- Acid secretion can be suppressed with drugs that block H2 receptors, proton pump inhibitors.
- Anticholinergic drugs may be used to inhibit gastric secretion, suppress gastric motility, and delay gastric emptying. (Physiology: Through vagal stimulation, acetylcholine stimulate pepsin and acid secretion.

Aggressive factors:

Pepsin
Acid (also stimulates chief cells to secrete pepsin
H. Pylori
Nicotine
Alcohol
Drugs (NSAIDs)
Delayed stomach emptying
Stress

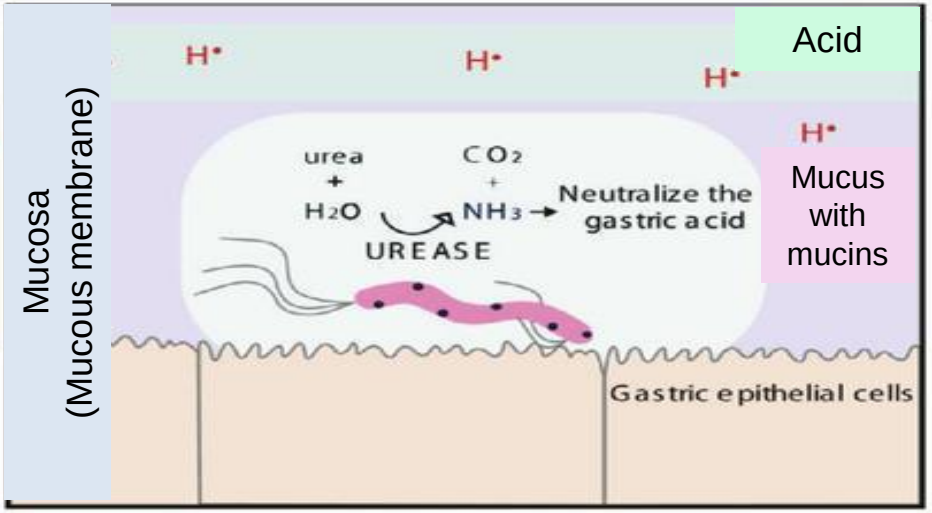
Peptic ulcer: Helicobacter Pylori

1. A characteristics feature: the synthesis of urease
It catalyzes urea hydrolysis with the formation of ammonia (NH_3) and carbon dioxide.
By this mechanism,

- H. pylori neutralizes the surrounding gastric acid and protects itself from the strong acidity of the stomach ($NH_3 + H^+ = NH_4^+$ is less acidic than H^+)
- Metabolites from urease activity are toxic to gastric epithelial cells: The formed ammonium reacts with hypochloride anion (OCl^-) produced by activated neutrophils to form highly toxic monochloramine (NH_2Cl) in the stomach, a hallmark of H. pylori infection.

2. Spiral shape and the motility of their flagella: render them resistant to peristaltic flushing of the gastric contents and enable them to persist in the mucous layer.

3. Pathogen produces other enzymes including catalase, oxidase, protease, and phospholipase, as well as it synthesizes specific adhesion proteins that enable them to adhere to mucous and epithelial cells



Helicobacter Pylori

Induced inflammation and ROS production

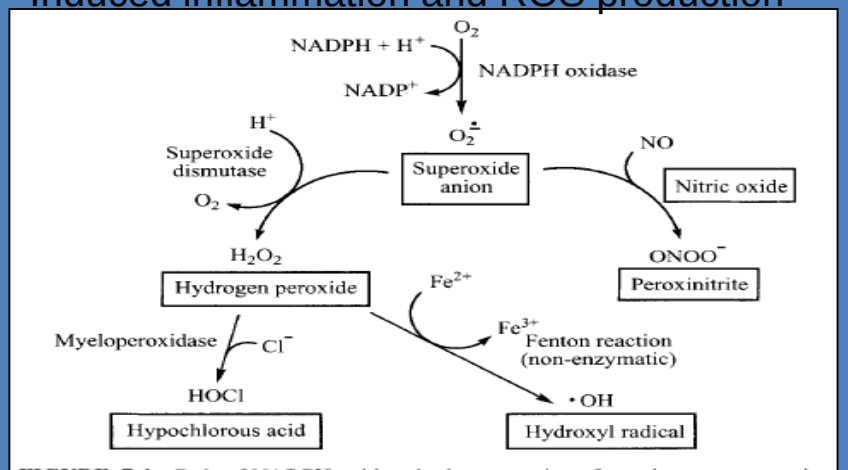


FIGURE 7-1 Role of NADPH oxidase in the generation of reactive oxygen species.

- Enterochromaffin-like cells secrete histamine. Histamine receptors in the gastric mucosa are H2 receptors (unlike those in the bronchial mucosa, which are H1 receptors). They stimulates gastric acid secretion.

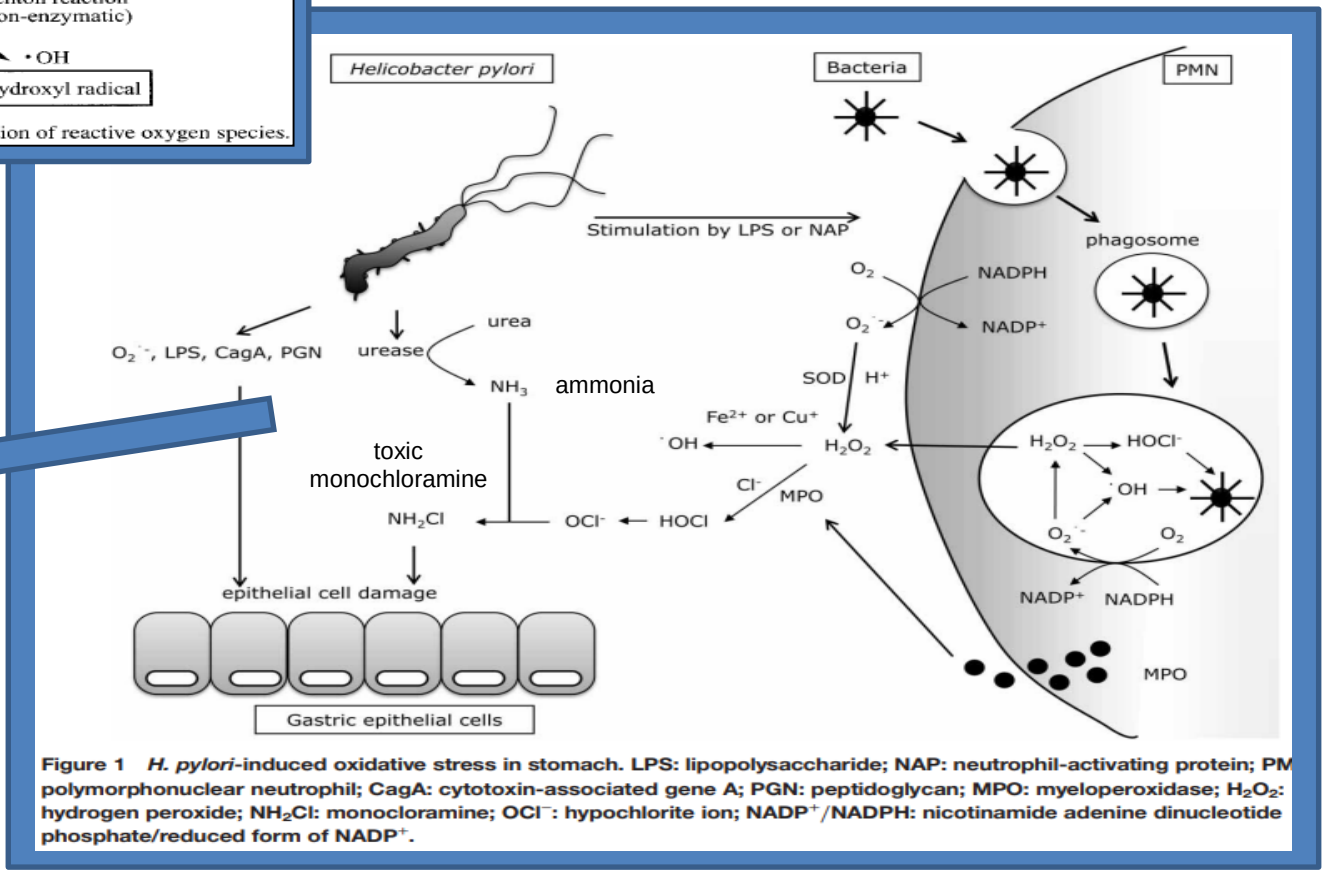


Figure 1 ***H. pylori*-induced oxidative stress in stomach. LPS: lipopolysaccharide; NAP: neutrophil-activating protein; PM polymorphonuclear neutrophil; CagA: cytotoxin-associated gene A; PGN: peptidoglycan; MPO: myeloperoxidase; H_2O_2: hydrogen peroxide; NH_2Cl: monocloramine; OCl^-: hypochlorite ion; $NADP^+$/NADPH: nicotinamide adenine dinucleotide phosphate/reduced form of $NADP^+$.**

Gastric ulcers

Stress ulcer or Stress-Related Mucosal Disease

- An acute form of peptic ulcer
- Cause: the physiologic stress of severe illness; multisystem organ failure; or major trauma, including severe burns or head injury. Usually, multiple sites of ulceration are distributed within the stomach or duodenum.
- Stress ulcers may be classified as:
  - **Ischemic ulcers**: within hours of an event(hemorrhage, multisystem trauma, severe burns, heart failure, or sepsis) that causes ischemia of the stomach and duodenal mucosa. Stress ulcers that develop as a result of burn injury are often called **Curling ulcers.**
  - **Cushing ulcers:** named after Harvey Cushing, is a gastric ulcer associated with elevated intracranial pressure. Cushing ulcer results from decreased mucosal blood flow and hypersecretion of HCL caused by overstimulation of the vagal nuclei after severe head trauma or brain surgery.

Postprandial Dumping syndrome

rapid emptying of hypertonic chyme from the surgically created, residual stomach into the small intestine 10 to 20 minutes after eating

- After partial gastrectomy, bariatric surgical procedures, or pyloroplasty

The small intestine: content

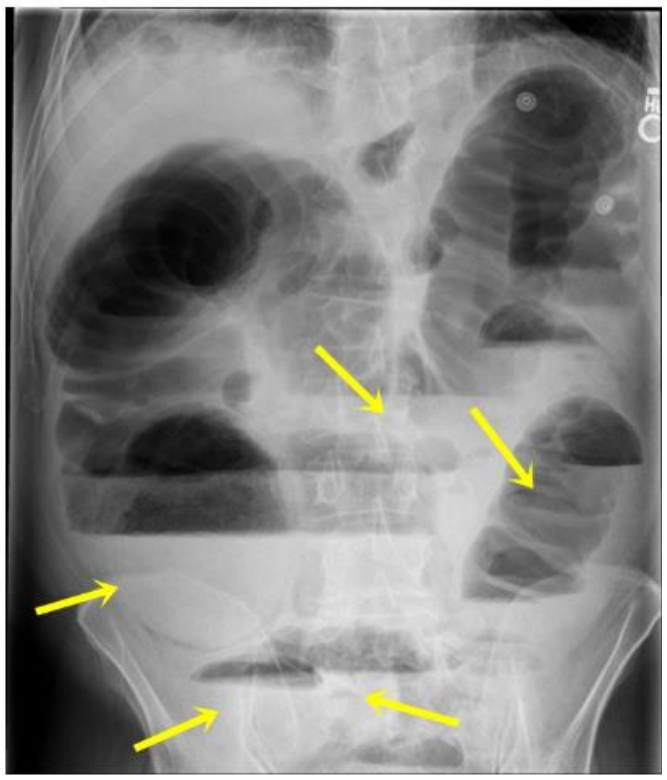
- Physiological review: Mucosa of the small intestine
- Malassimilation: primary and secondary
- The intestine obstruction or ileus
- Inflammatory diseases: Crohn disease

Mucosa of the small intestine

Hormone	ACTION
Motilin	• Increases gastrointestinal motility
Secretin	• Stimulates alkaline pancreatic juice and liver to secrete bile • Decreases gastrointestinal motility (similar to glucagon and adrenalin) • Inhibits gastrin and gastric acid secretion
Cholecystokinin	• Stimulates gallbladder to eject bile (contraction) and pancreas to secrete alkaline fluid • Decreases gastric motility and constricts pyloric sphincter and increased intestinal motility (similar to insulin) • Inhibits gastrin; delays gastric emptying
Enterokinase (enteropeptidase)	• the pancreatic enzymes are activated by enterokinase that is secreted in duodenum
Incretins (e.g. glucagon-like peptide from L cells)	• Increase sensitivity of pancreatic beta cells of the islets of Langerhans and thereby increasing insulin release (therapy in Type II DM) • Inhibits glucagon release from the alpha cells of the islets of Langerhans.

Small intestine obstruction: general clinical manifestation

- Vomiting: initially arising reflexively, later from accumulation of contents
- Abdominal pain (colicky, sharp pain arising from distension)
- Constipation and meteorism or tympanites (flatulence and abdominal bloating due to increased intestinal gas)
- Abdominal swelling (we describe it as " *belly above niveau* ").



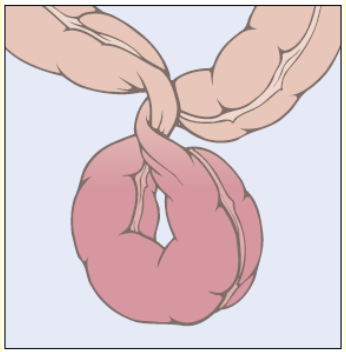
X-ray image of the abdomen showing hydroaeric phenomenon: horizontal layers of intestinal contents (white) and above them dilated intestinal villi filled with gas (dark areas)

Small intestine obstruction: pathophysiology

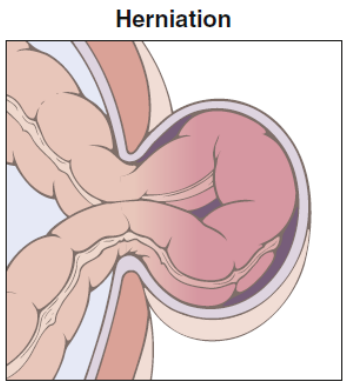
- Distention (pain: colicky character) begins almost immediately due to accumulation of fluid and gas inside the lumen proximal to the obstruction.
    - Fluids accumulate: within 24 hours, up to 8 L of fluid enters the lumen from
        - enhanced movement of fluid from the vascular space to the intestinal lumen
        - impaired water and electrolyte absorption.
        - saliva, gastric juice, bile, pancreatic juice, and intestinal secretions
    - Gas from swallowed air, and to a lesser extent from bacterial overgrowth, contributes to the distention.
  - Distention decreases the intestine's ability to absorb water and electrolytes and increases the net secretion of these substances into the lumen.
    - Compared to the proximal small intestine obstruction in the distal small intestine causes more pronounced distention because a greater length of intestine is proximal to the obstruction
  - Distention push against the diaphragm: decrease lung volume leading to atelectasis and pneumonia (particularly in debilitated individuals)
  - Extracellular fluid volume and plasma volume decrease, causing dehydration, elevated hematocrit, hypotension and tachycardia. Severe dehydration leads to hypovolemic shock.
- Copious vomiting ($\downarrow K^+$, $\downarrow Na^+$)
  - Obstruction in the proximal small intestine causes mild distention and vomiting of bile-stained fluid.
  - Obstruction in the distal small intestine vomiting may not occur or may occur later and contain fecal material.
- Electrolyte disturbances:
  - Pyloric or higher obstruction: metabolic alkalosis
    - Excessive loss of hydrogen ions that normally would be reabsorbed from the gastric juice
  - Prolonged obstruction or obstruction lower in the intestine: metabolic acidosis
    - bicarbonate from pancreatic secretions and bile cannot be reabsorbed
    - ketosis result from starvation. Lack of circulation permits
    - lactic acid
- Occluded arterial circulation: ischemia, necrosis, perforation, and peritonitis, overgrowth of bacteria (fever, leukocytosis) or sepsis.

Small intestine obstruction: types of ileus

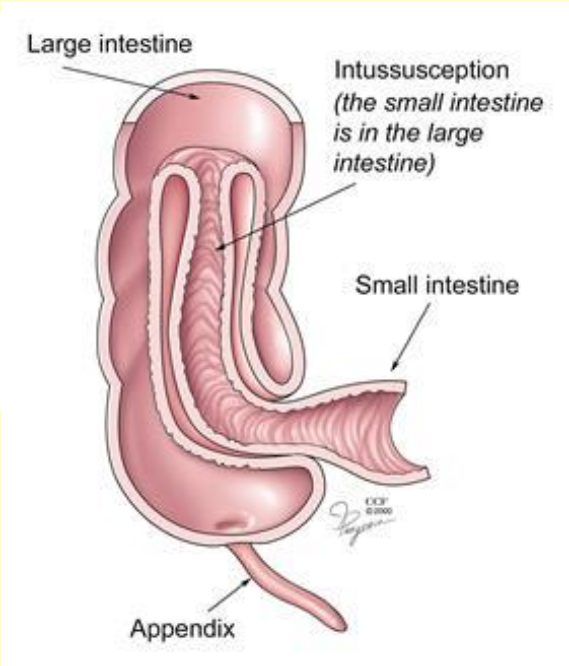
- **Mechanical**:
  - Obturation: Impacted stool
  - Volvulus: Disorder of intestinal rotation or fastening to the back wall of the abdominal.
  - Strangulation: external obstruction
- **Neurogenic**
  - Paralytic ileus
  - Spastic
  - Mixed
- **Vascular:** due to arterial embolism or venous thrombosis



Volvulus







Small intestine obstruction: Simple mechanical ileus (obstructive)

- It is caused by the impacted stool
- Manifestation:
  - **Colicky pains:** not very strong, the intensity gradually increases and later permanent pain from distension with a decrease in peristalsis.
  - **Vomiting**: occurs later due to stagnation of bowel contents above the obstruction. The more distal the obstruction, the later it occurs.)
  - **Tactile soreness:** it is mainly localized in the area of the obstacle. After moving the palpating hand away, the pain disappears
  - **Peristaltic waves:** visible on the abdominal wall and audible effortful peristalsis



Small intestine obstruction: Strangulation ileus

- It is caused by external obstruction to the small bowel, which results in ischemia and loss of bowel peristalsis
  - Causes: adhesions in the abdomen that form after surgery; hernias; colon cancer; certain medications; or strictures from an inflamed intestine caused by Crohn's disease or diverticulitis

- Manifestation
  - **Colicky pains:** very strong from the beginning
  - **Tactile soreness:** in the place of the obstacle, does not disappear when the examining hand is moved away.
  - **Vomiting:** from the beginning
  - **Flatulence:** with stoppage of passing gases and stools; troubles the patient from the beginning of the episode.
  - A mixture of blood and mucus in the stool.
  - Disappearance of peristalsis.

A drop in blood pressure, an increase in heart rate, pallor and even cyanosis of the face .

Small intestine obstruction: Paralytic ileus ="**dead silence„**

- Causes: post-operative conditions, intoxication, etc.
- Manifestation
  - **Weaker pains** from bowel and abdominal wall distension, colic pains are not present.
  - **Gases and stools do not pass and t**he abdomen is **evenly inflated.**
  - **The patient is in a relatively good condition: normal** blood pressure, heart rate, body

Small intestine obstruction: Vascular ileus

- Causes: arterial embolism or venous thrombosis, we are interested in heart or vascular diseases.
- Manifestation
  - **Sharp to shocking pain**.
  - From the beginning **reflex vomiting**.
  - **Loose stool with blood admixture** appears.
  - **Increase in heart rate, decrease in blood pressure**.

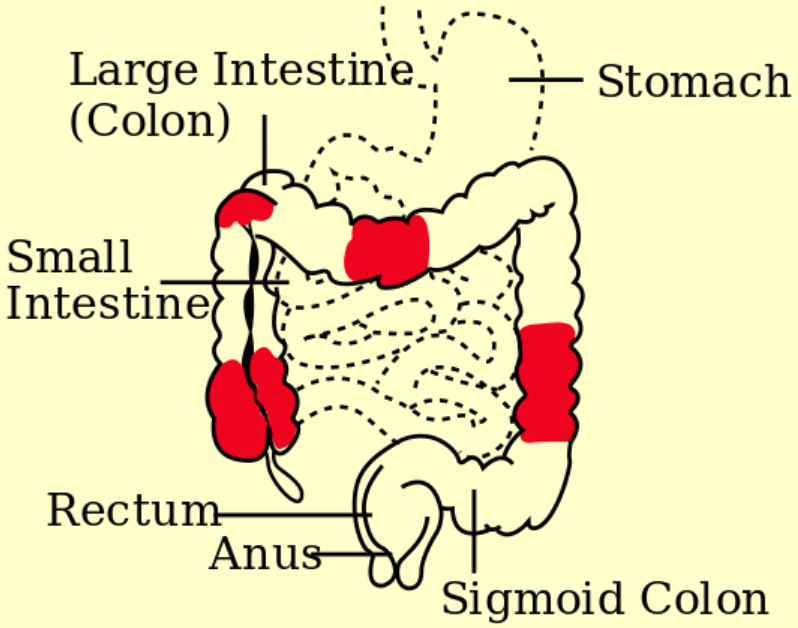
Small intestine obstruction:

Spastic ileus

Inflammatory diseases: Crohn disease

- CD affects any part of the gastrointestinal tract from the mouth to the anus. The distal small intestine (malabsorption) and proximal large intestine are most commonly affected (patchy distribution in the colon or ileum).
- The typical chronic lesion is a skip granuloma having cobblestone projections of inflamed tissue surrounded by areas of ulceration. It starts in the intestinal submucosa and spreads across the intestinal wall to involve the mucosa and serosa in areas overlying lymphoid tissue. Complications include obstruction, fistulae, abscess formation, and chronic blood loss.
- Th1 immune response and IgG associated with severity
- Its causes are unknown: aggressive response to normal flora bacteria in genetically predisposed individuals (e.g. TLR)

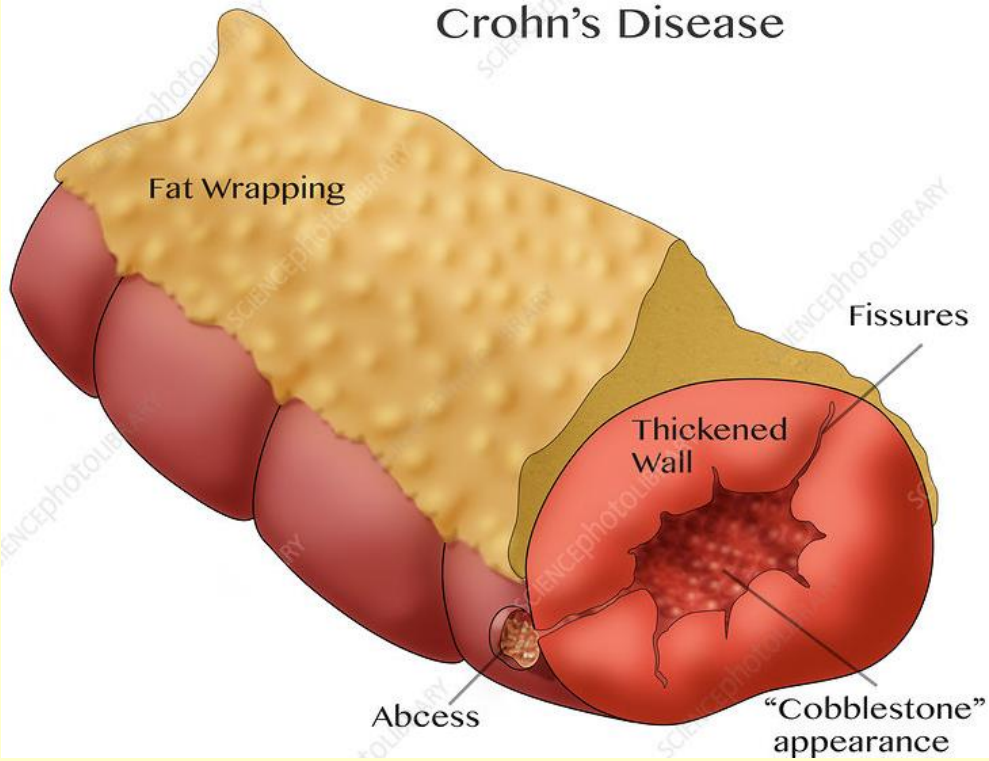
Inflammatory diseases: Crohn disease



Crohn's Disease

The distal small intestine (malabsorption) and proximal large intestine are most commonly affected (patchy distribution in the colon or ileum)

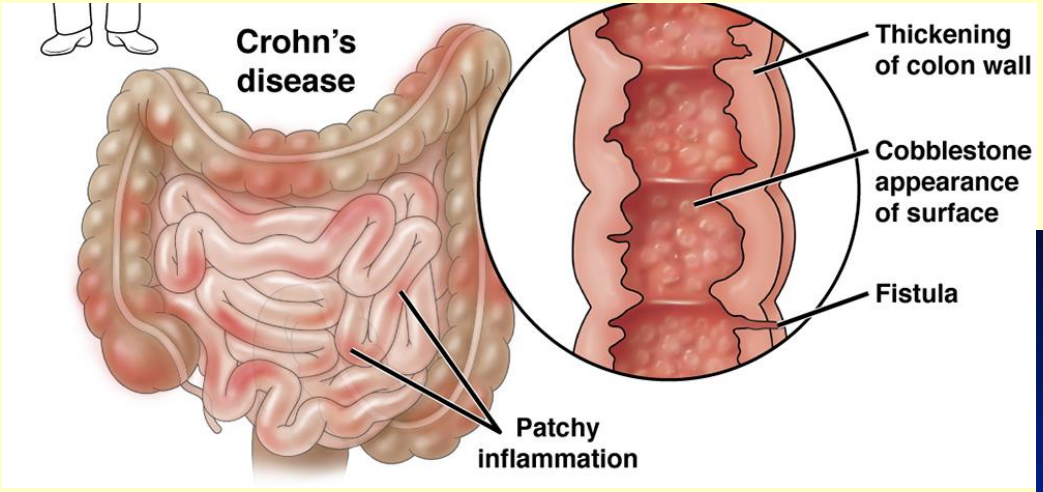
Inflammatory diseases: Crohn disease



It starts in the intestinal submucosa and spreads across the intestinal wall to involve the mucosa and serosa in areas overlying lymphoid tissue. Complications include obstruction, fistulae, abscess formation, and chronic blood loss.

Inflammatory diseases: Crohn disease







Complications include obstruction, fistulae, abscess formation, and chronic blood loss.

- The affected serosis leads to adhesions in which fistulas are formed.
- The fibrosis leads to stenosis and colicky pain (subileus state)

The large intestine

- Diverticular Disease of the Colon
- Colon polyps: cancer
- Megacolon
- Irritable Bowel Syndrome
- Inflammatory diseases: Ulcerative colitis

Diverticular Disease of the Colon

Diverticula: herniation or saclike outpouchings of mucosa through the muscle layers of the colon wall.

- anywhere in the gastrointestinal tract but the most common site is the left colon. They rarely occur in the small intestine
- The diverticula form at weak points in the colon wall, usually where arteries penetrate the tunica muscularis to nourish the mucosal layer.
- A common associated finding is thickening of the circular and longitudinal (teniae coli) muscles surrounding the diverticula. Hypertrophy and contraction of these muscles increase intraluminal pressure and degree of herniation.

Diverticulosis: asymptomatic diverticular disease.
Diverticulitis represents inflammation.

Causes: unknown but is associated with age greater than 60 years, combination of decreased dietary fiber, increased intracolonic pressure, abnormal neuromuscular function, and alterations in intestinal motility

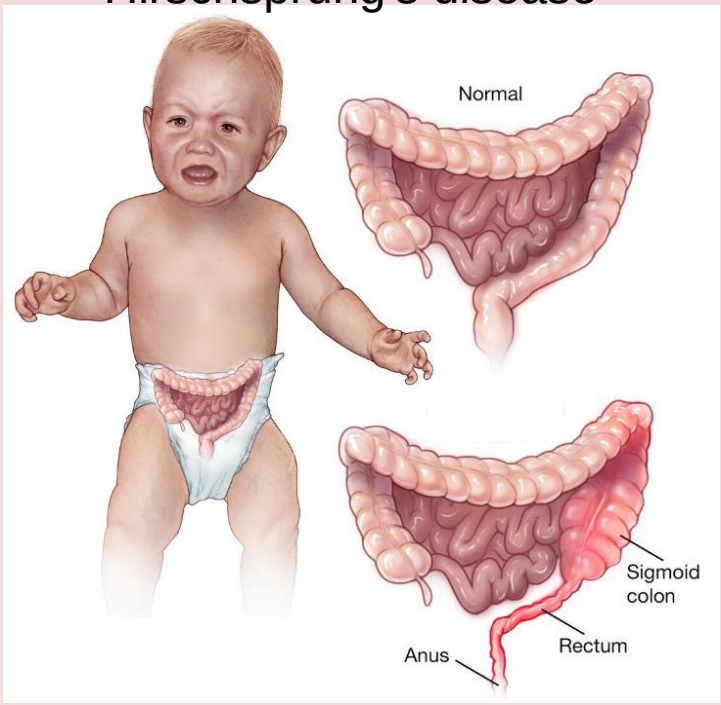
Colon polyps

- Non-neoplastic polyps: hyperplastic polyps, or inflammatory polyps
- Neoplastic polyps (epithelial dysplasia or abnormal cells): adenomas (benign) and carcinomas (malignant): colorectal carcinoma

Megacolon

- An abnormal dilation of the colon
  - Congenital or aganglionic megacolon or Hirschsprung's disease: the ganglion cells of the myenteric plexus are absent.
  - Acquired megacolon: Parkinson diseases
  - Toxic megacolon in ulcerative colitis due to excessive production of nitric oxide colitis.

Hirschsprung's disease



Acute colonic pseudo-obstruction (Ogilvie syndrome)

- a massive dilation of the large bowel (the cecum) that occurs in critically ill patients and immobilized older adults.
- cause: excessive sympathetic motor input or decreased parasympathetic motor input.
- absence of mechanical obstruction

Irritable Bowel Syndrome

- Abdominal pain and altered bowel habits (diarrhea-predominant, constipation-predominant)
- Common in women, with a higher prevalence in youth and middle age.
- Causes: anxiety and depression
- Pathophysiology:
  - *Visceral hypersensitivity or hyperalgesia,* particularly with distention of the rectum but also other areas of the gut.
  - *Abnormal gastrointestinal permeability, motility, and secretion:* The mechanisms may be related dysregulation of the brain-gut axis or to the role of serotonin
  - *Overgrowth of intestinal flora*
  - *Food allergy or food intolerance*
  - *Psychosocial factors*

Ulcerative colitis (UC)

- **UC** is a chronic inflammatory disease that causes ulceration of the colonic mucosa and extends proximally from the rectum into the colon: continuous lesions, are limited to the mucosa, and are not transmural. The mucous layer is thinner than normal bud edema and thickening of the muscular mucosae may narrow the lumen of the involved colon.
- In chronic disease, inflammatory polyps (pseudopolyps) develop in the colon from rapidly regenerating epithelium(also in CD). Mucosal destruction causes bloody diarrhea with passage of purulent mucus but severe hemorrhage is rare.
- T cells, plasma cells in the inflamed colon and colonic epithelial antibodies IgG in the circulation
- Its causes are unknown: bacterial, genetic, immunologic or dietary factors

References

Look up the particular theme in the following indexes

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